Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/02/2024
PI Name:	Doherty, Colleen Ph.D.		
Project Title:	How Do Carbon Fixing Strategies Affect Nutritional Content Under High CO2? A Comparison of C3 vs. C4 Microgreens		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27695-7622	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	12/08/2022	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 12/17/24). NOTE: End date changed to 12/06/2024 per NSSC information (Ed., 10/9/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC23K0344		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several carbon capture mechanisms have emerged in plant systems that provide unique advantages to plants depending on their environment. For example, while most plants use C3 photosynthesis, C4 and Crassulacean acid metabolism (CAM) carbon capture mechanisms can increase water use efficiency or temperature tolerance. These advantages have been well-characterized in the atmospheric carbon dioxide (CO_2) levels on Earth, but in enclosed human habitats such as those needed for long-term spaceflight, CO_2 levels far exceed that of the Earth's atmosphere. Altered CO_2 levels are known to affect nutritional content and water use efficiency; however, much of the prior research has used CO_2 levels below that on enclosed human habitats. This proposed work would examine how high CO_2 levels affect plant physiology and nutritional content of edible microgreens that use different photosynthetic mechanisms: C3, C4, and CAM. We will monitor physiological characteristics and the nutritional profile across different CO_2 levels for select microgreen species (with C3, C4, and CAM photosynthesis). The combined effects of altered CO_2 levels and other spaceflight relevant stresses such as water availability will be examined to understand if these different photosynthetic mechanisms can provide advantages to enhance plant productivity in space environments. These results would provide important baseline information on plant nutrition and performance that is needed for planning long-term space missions and thus would address the following objectives of the solicitation and NASA program goals: Decadal Survey-Priority 3: A systematic suite of plant biology experiments to elucidate mechanisms by which plants respond and adapt to spaceflight, and to facilitate their eventual use in Bioregenerative Life Support Systems; PB-1 How does gravity affect plant growth, development & metabolism (e.g. photosynthesis, reproduction, lignin formation, plant defense mechanisms); and PB-3 How can horticultural approaches for sustained production of edible crops in space be both improved and implemented (especially as related to water and nutrient provision in the root zone)?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	CO_2 levels are increasing on Earth, which may impact plant physiology. However, extreme CO_2 levels, as are found in contained human environments such as on the ISS, are not commonly found in plant-growing areas on Earth. Understanding how plants perform in these environments and the effects on physiology and nutrition could affect decisions about using such environments to increase crop production on Earth.
Task Progress:	The construction of the eCO_2 chamber has been completed and optimized. Plants can now be grown in ambient or eCO_2 with consistent humidity in the chamber. Microgreens for both C3 and C4 plants have been grown in the chamber with multiple replicate trails, switching chambers to control for chamber-specific variation. Physiology and growth characteristics have been evaluated and the differences in the response to eCO_2 between the plants have been determined. We are currently writing this up and will present the findings at the American Society for Gravitational and Space Research (ASGSR). The evaluation of the effects of eCO_2 on nutritional content is ongoing. The pipeline for the Nuclear Magnetic Resonance (NMR) has been established; sample preparation protocols have been demonstrated for each microgreen species. Samples collected from each experiment are being processed for NMR analysis. This part has not yet been completed and will extend a few more months. In part, this delay is due to multiple power outages. We anticipate that processing these samples should be complete in a few months and the analysis shortly after that. The combined physiology, growth, and metabolite analysis will be prepared for publication.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Toennisson A, Horton TB, Perera IY, Doherty CJ. "Evaluating microgreen growth and nutrient content under elevated CO_2." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023